

pgAdmin 4

File Object Tools Edit View Window Help

Object Explorer

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Query

```

SELECT * FROM livro

SELECT
a.id_autor,
a.nome,
(
SELECT COUNT(*)
FROM livro l
WHERE l.id_autor = a.id_autor
) AS quantidade_livros,
(
SELECT ROUND(AVG(l.num_paginas), 2)
FROM livro l
WHERE l.id_autor = a.id_autor
) AS media_paginas
FROM autor a
ORDER BY a.nome;

```

Query History

Scratch Pad

Data Output

Showing rows: 1 to 4 Page No: 1 of 1

id_autor [PK] integer	nome character varying (100)	quantidade_livros bigint	media_paginas numeric
3	Clarice Lispector	1	88.00
1	J. R. R. Tolkien	2	952.00
4	J.K. Rowling	0	[null]
2	Machado de Assis	1	208.00

Total rows: 4 Query complete 00:00:00.105 CRLF Ln 109, Col 19

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Query

```

WITH livros_por_autor AS (
SELECT
id_autor,
COUNT(id_livro) AS quantidade_livros,
ROUND(AVG(num_paginas), 2) AS media_paginas
FROM livro
GROUP BY id_autor
)
SELECT
a.id_autor,
a.nome,
lpa.quantidade_livros,
lpa.media_paginas
FROM autor a
LEFT JOIN livros_por_autor lpa
ON lpa.id_autor = a.id_autor
ORDER BY a.nome;

```

Query History

Scratch Pad

Data Output

Showing rows: 1 to 4 Page No: 1 of 1

id_autor [PK] integer	nome character varying (100)	quantidade_livros bigint	media_paginas numeric
3	Clarice Lispector	1	88.00
1	J. R. R. Tolkien	2	952.00
4	J.K. Rowling	[null]	[null]
2	Machado de Assis	1	208.00

Total rows: 4 Query complete 00:00:00.100 CRLF Ln 116, Col 1

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Query

```
136 WITH paginas_por_autor AS (  
137 SELECT  
138 id_autor,  
139 SUM(num_paginas) AS total_paginas  
140 FROM livro  
141 GROUP BY id_autor  
142 )  
143 SELECT  
144 a.id_autor,  
145 a.nome,  
146 ppa.total_paginas  
147 FROM autor a  
148 JOIN paginas_por_autor ppa  
149 ON ppa.id_autor = a.id_autor  
150 WHERE ppa.total_paginas > (  
151 SELECT AVG(num_paginas)  
152 FROM livro  
153 )  
154 ORDER BY ppa.total_paginas DESC;
```

Query History

Scratch Pad

Data Output Messages Notifications

Showing rows: 1 to 1 Page No: 1 of 1

id_autor [PK] integer	nome character varying (100)	total_paginas bigint
1	J. R. R. Tolkien	1904

Total rows: 1 Query complete 00:00:00.086 CRLF Ln 136, Col 1

Pesquisar

16:54 16/06/2026

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 - Casts
 - Catalogs
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 - Extensions
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Query

```
155  
156  
157 SELECT  
158 a.id_autor,  
159 a.nome,  
160 (  
161 SELECT SUM(l.num_paginas)  
162 FROM livro l  
163 WHERE l.id_autor = a.id_autor  
164 ) AS total_paginas  
165 FROM autor a  
166 WHERE (  
167 SELECT SUM(l.num_paginas)  
168 FROM livro l  
169 WHERE l.id_autor = a.id_autor  
170 ) > (  
171 SELECT AVG(num_paginas)  
172 FROM livro  
173 )  
174 ORDER BY total_paginas DESC;
```

Query History

Scratch Pad

Data Output Messages Notifications

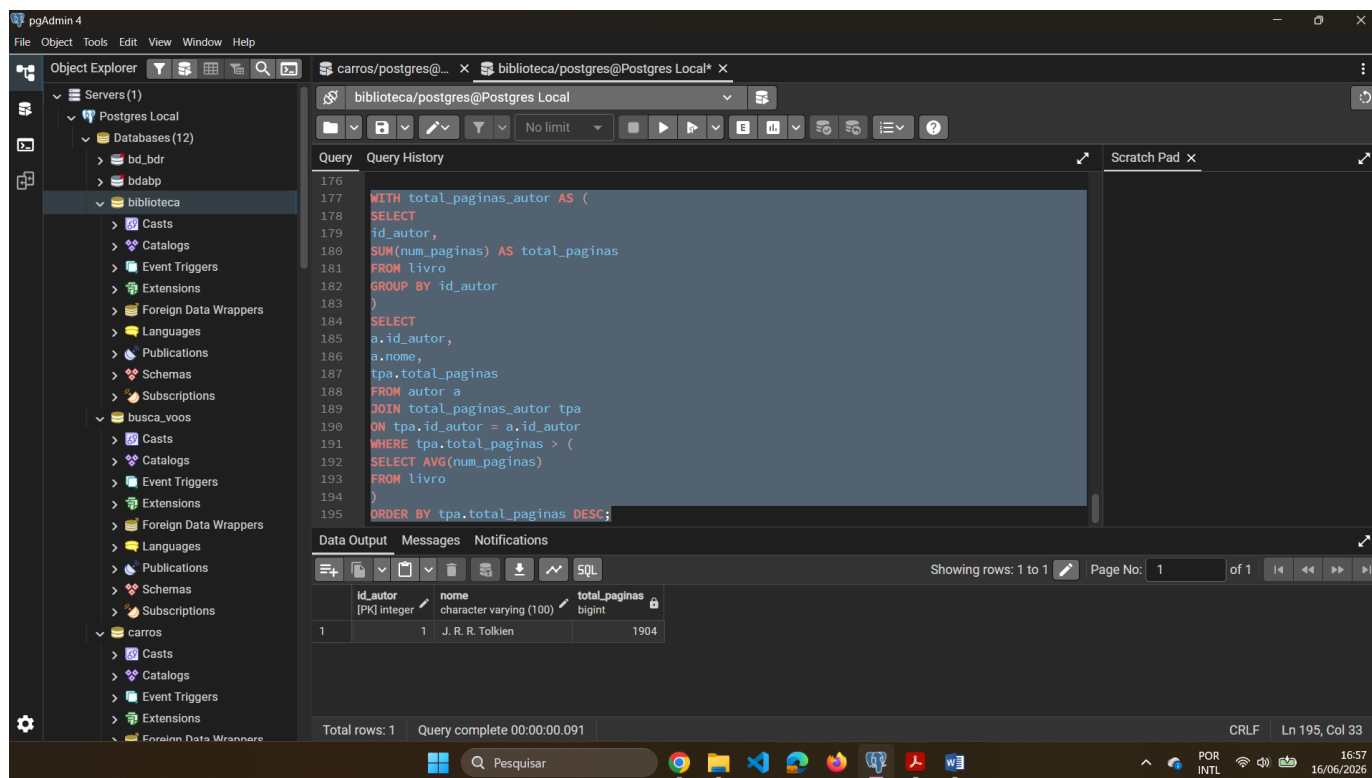
Showing rows: 1 to 1 Page No: 1 of 1

id_autor [PK] integer	nome character varying (100)	total_paginas bigint
1	J. R. R. Tolkien	1904

Total rows: 1 Query complete 00:00:00.081 CRLF Ln 157, Col 1

Pesquisar

16:56 16/06/2026



Marcus Vinicius Ribeiro do Nascimento

-- EXERCÍCIO 1 - Subquery scalar no SELECT

SELECT

a.id_autor,

a.nome,

(

SELECT COUNT(*)

FROM livro l

WHERE l.id_autor = a.id_autor

) AS quantidade_livros,

(

SELECT ROUND(AVG(l.num_paginas), 2)

FROM livro l

WHERE l.id_autor = a.id_autor

```
    ) AS media_paginas  
FROM autor a  
ORDER BY a.nome;
```

-- EXERCÍCIO 1 - Versão com CTE

```
WITH livros_por_autor AS (  
    SELECT  
        id_autor,  
        COUNT(id_livro) AS quantidade_livros,  
        ROUND(AVG(num_paginas), 2) AS media_paginas  
    FROM livro  
    GROUP BY id_autor  
)  
SELECT  
    a.id_autor,  
    a.nome,  
    lpa.quantidade_livros,  
    lpa.media_paginas  
FROM autor a  
LEFT JOIN livros_por_autor lpa  
    ON lpa.id_autor = a.id_autor  
ORDER BY a.nome;
```

-- EXERCÍCIO 2

```
WITH paginas_por_autor AS (  
    SELECT  
        id_autor,
```

```

        SUM(num_paginas) AS total_paginas
FROM livro
GROUP BY id_autor
)
SELECT
    a.id_autor,
    a.nome,
    ppa.total_paginas
FROM autor a
JOIN paginas_por_autor ppa
    ON ppa.id_autor = a.id_autor
WHERE ppa.total_paginas > (
    SELECT AVG(num_paginas)
    FROM livro
)
ORDER BY ppa.total_paginas DESC;

```

-- EXERCÍCIO 3 - A: Subconsulta correlacionada

```

SELECT
    a.id_autor,
    a.nome,
    (
        SELECT SUM(l.num_paginas)
        FROM livro l
        WHERE l.id_autor = a.id_autor
    ) AS total_paginas
FROM autor a
WHERE (
    SELECT SUM(l.num_paginas)

```

```
FROM livro l
WHERE l.id_autor = a.id_autor
) > (
SELECT AVG(num_paginas)
FROM livro
)
ORDER BY total_paginas DESC;
```

-- EXERCÍCIO 3 - B: CTE pré-agrupada

```
WITH total_paginas_autor AS (
SELECT
    id_autor,
    SUM(num_paginas) AS total_paginas
FROM livro
GROUP BY id_autor
)
SELECT
    a.id_autor,
    a.nome,
    tpa.total_paginas
FROM autor a
JOIN total_paginas_autor tpa
    ON tpa.id_autor = a.id_autor
WHERE tpa.total_paginas > (
    SELECT AVG(num_paginas)
    FROM livro
)
ORDER BY tpa.total_paginas DESC;
```